Executive Summary: Delivery Performance, Delay Risk, and Logistics Efficiency Analysis in Global Supply Chain Operations

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1. Business Context

APL has a lot of shipment data sitting around: delivery times, shipping modes, regions, customers, products, sales and profit. Most of it only gets used to count what already happened. This project was about looking closer: could the same records show where deliveries run late, how efficiently shipments move, and whether late deliveries line up with weaker financial results?

2. Business Problem

Right now, there's no structured way to see:

1. How actual shipping time compares with the schedule.
2. Which groups of shipments carry the most delays.
3. Whether recorded profit differs when deliveries are late.

Without that, decisions about where to investigate end up leaning on overall totals, which hide a lot. This project builds a cleaned dataset, a set of delay and efficiency measures and an interactive dashboard to give those decisions something more concrete to work from.

3. Project Objectives

- Compare actual and scheduled shipping duration.
- Understand how delivery outcomes and the risk label are distributed.
- Compare performance across shipping modes, regions, markets, countries, customer segments and products.
- Check whether recorded financial results differ across delay groups.
- Judge data quality properly, not just by eye.
- Package the results into a reusable set of KPIs and analysis tables.
- Put it behind an interface non-technical people can use.
- Turn the numbers into practical recommendations.
- Make sure the dashboard doesn't break on bad input.

4. Data Overview

The source file is APL_Logistics.csv.

Item	Value
Shipment records	180,519
Original columns	40

Item	Value
Columns at Day 19 diagnostics	54
Initial missing cells	11
Duplicate rows	0
Date column	Not available

The first check found 8 missing surnames and 3 missing postcodes, which were filled with the value Unknown. The original file was left untouched and all work was done in separate files.

5. Key Delivery Measures

Late_delivery_risk

98,977 shipments (about 54.83%) carry a high-risk label, and 81,542 are low risk. That describes what's recorded in the dataset. It isn't a prediction.

Delay duration

Group	Shipments
No Delay	77,119
1–2 Days	89,365
3–5 Days	14,035

The three groups add up to all 180,519 records, and the longest delay seen was four days.

6. Analytical Methodology

The work moved through the usual stages in order: cleaning, feature engineering, exploring the data, comparing actual and scheduled durations, delay classification, financial comparison, KPI development, diagnostic validation, and finally a Streamlit dashboard with testing at the end.

7. Delay and Financial Analysis

Average recorded profit per shipment falls as delays get longer:

Delay Duration	Shipments	Total Recorded Profit	Average Recorded Profit
No Delay	77,119	1,734,936.32	22.50
1–2 Days	89,365	1,940,779.93	21.72
3–5 Days	14,035	291,186.72	20.75

Amounts are in the dataset's recorded monetary units.

8. Key Analytical Observation

This is probably the most important finding in the whole project: a metric is only as trustworthy as its definition.

The efficiency ratio I built divides by actual shipping duration, and 5,080 values came out undefined because actual duration was zero on those records. I kept them undefined instead of forcing them to zero.

The same care applies to the risk label. It records an outcome, and it shouldn't be read as a forecast. Fields built from completed deliveries, such as the delay flags, would leak the answer if anyone used them to predict a delay in advance.

9. Interactive Dashboard

The final solution lives in a Streamlit app (`app\app.py`). It has an executive overview and nine analytical modules:

- Delivery Performance
- Shipping Efficiency
- Financial Performance
- Regional Analysis
- Market Analysis
- Country Analysis
- Customer Segments
- Product & Category
- Relationship Analysis

The regional, market, segment and product comparisons live in these modules and aren't reproduced in this summary.

10. Business Value

Delay monitoring — tracking counts and rates together shows where delays concentrate and where they just reflect volume.

Shipping-mode review — comparing actual and scheduled durations by mode points to where to look first.

Geographical review — regional, market and country views help prioritise operational reviews.

Financial context — profit by delay group is useful context, though it shouldn't be read as proof that delays cause lower profit.

Accessibility — none of this is useful if it stays in a notebook, so the dashboard exists to put it in front of non-technical decision-makers.

11. Data Quality and Diagnostics

The Day 19 diagnostics reported 180,519 records, 54 columns, no duplicates and 5,080 missing cells, nearly all in the efficiency ratio.

One item still needs explaining: the 1–2 day and 3–5 day groups together total 103,400 delayed shipments, while 98,977 are labelled high risk, a gap of 4,423. The two measure different things, but I haven't yet traced the difference.

12. Dashboard Testing and Reliability

During development, a missing `statsmodels` dependency caused errors in two charts, which I fixed by removing the trendline setting. I also defined a 12-point filter-testing checklist, covering default loading, single and multiple selections, resets, KPI updates and empty results.

I want to be plain about this: the executed test results aren't documented yet. Passing code and a cleared error are not the same as a completed test, so I'm not claiming a pass rate.

13. Key Results

Area	Result
Shipment records analysed	180,519
High-risk labelled shipments	98,977
Delay groups	3
Undefined efficiency-ratio values	5,080
Dashboard modules	10 sections
Interactive dashboard	Streamlit
Dashboard filter testing	Defined, results not yet documented

Worth calling out plainly: the profit pattern is descriptive. It shows that average profit falls with longer delays, not that delays cause it.

14. Business Recommendations

1. Track delayed-shipment counts and delay rates together.
2. Compare shipping modes using volumes, actual durations and scheduled durations.
3. Investigate the large 1–2 day delay group first.
4. Review the 3–5 day group for recurring route or scheduling problems.
5. Look at profit alongside discounts, product mix and shipment characteristics before blaming delays.
6. Keep one written definition each for a delay and for late-delivery risk.
7. Finish and document the dashboard testing before relying on it.
8. Add reliable order or shipment dates to unlock trend analysis.

15. Limitations

The biggest limitation is the missing date column, which rules out trends and seasonality. Beyond that, the data is historical and observational, so no relationship here proves cause and effect, and the risk label isn't a validated predictive model. The dashboard testing is also unfinished, so its operational readiness is unconfirmed.

16. Future Scope

- Add a validated order or shipment date.
- Bring in routes, carriers, service levels and transport costs.
- Compare profit across delay groups while controlling for order value, category, discounts and market.
- Build a properly validated prediction model if forward-looking estimates are needed.

- Complete and document filter and runtime testing.
- Automate data-quality checks as new data arrives.

17. Executive Conclusion

This project built a working, end-to-end analysis of APL's shipment data and packaged it into a dashboard that turns a large dataset into something a business user can explore.

The honest headline is that the findings are useful but not yet final. Delays and lower average profit move together, 98,977 shipments are labelled high risk, and the efficiency ratio needed careful handling, but the dashboard testing and the delayed-versus-high-risk gap still need closing.

That's a useful, specific result on its own, because it gives a clear direction for the next round of work. It gives APL a real foundation to build from as better data, especially dates, becomes available.

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